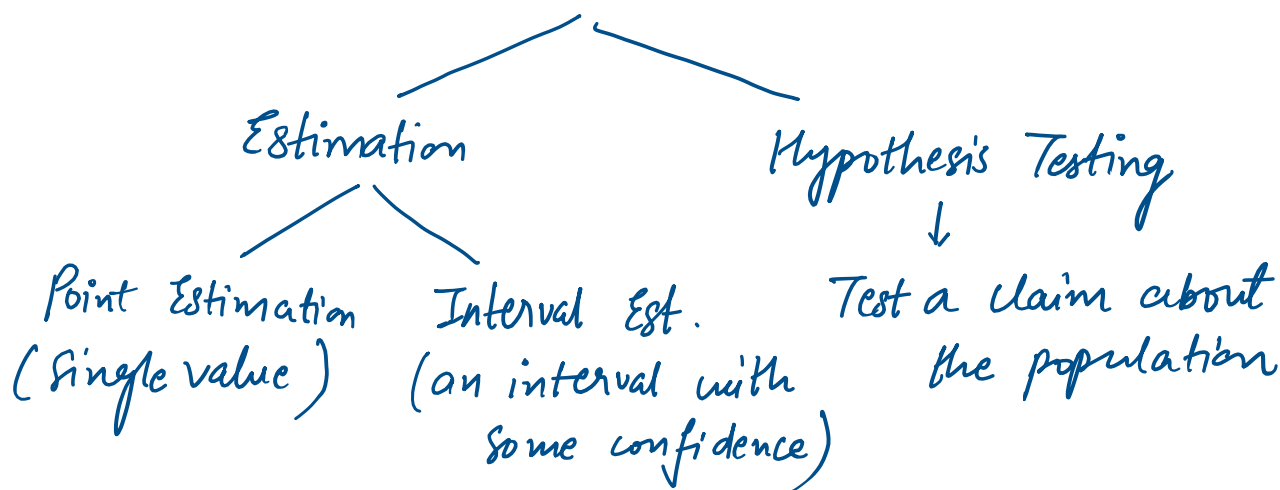


12/09/2026. Topics we have done :-

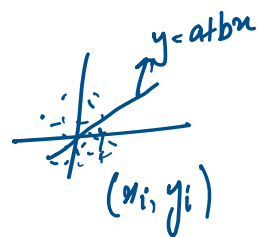
- ① Recap of probability
- ② Order statistics
- ③ Sampling distⁿ :- Gamma, chi-square, t and F distⁿ

Introduction to Estimation

Problem of Statistical Inference



Method of least squares :- It estimates the unknown parameters of a popⁿ by choosing their values so that the sum of squared differences between the observed and fitted values is minimum



for straight line model, $y = a + bx$

The objective is $\min_{a, b} \sum_{i=1}^n (y_i - a - b x_i)^2$

$$(x_i, y_i), i=1 \dots n$$
$$y = a + bx$$
$$\Rightarrow y_i = a + b x_i$$

Do yourself.

Def :- A parameter of a popⁿ is the characteristic which we are interested in.

Statistic
estimator

→ parameter
to be estimated

Point Estimation

Let $X_1, X_2, \dots, X_n$ be a random sample from a prob. distⁿ $f_X(x, \theta)$ where θ is the unknown parameter (or vector of unknown parameters). Then problem of point estimation is to provide an estimate for θ on the basis of $X_1, X_2, \dots, X_n$.

Estimator:- Any fⁿ of the random sample which is used to estimate the unknown parameter θ .

$$\bar{X} = \frac{X_1 + \dots + X_n}{n}, \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Thus if $\underline{X} = (X_1, \dots, X_n)$ is a random sample from a population with prob. distⁿ $f_X(x, \theta)$. Then a fⁿ $T(\underline{X}) = T(X_1, \dots, X_n)$ used to estimate θ or $g(\theta)$ is known as the estimator.

Suppose $\underline{x} = (x_1, x_2, \dots, x_n)$ be a realization of $\underline{X}$, then

$T(\underline{x})$ is called an estimate of θ or $g(\theta)$.

$g(\theta) / \theta \rightarrow$ to be estimated

$T(\underline{X}) \rightarrow$ estimator

Ex:- In estimating the average height of a group of people in a city, the sample mean $\bar{X}$ can be considered as an estimator of height. That is $T(\underline{X}) = \bar{X}$.

A random sample of 100 people from the city has sample mean 160 cm. $\Rightarrow T(\underline{x}) = 160$

Parameter space:- It is the set of all possible values of a parameter θ . It is denoted as $\Theta \rightarrow$ script theta.

Ex:- ① $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$

$$\theta \in \Theta, \quad \theta = (\mu, \sigma^2)$$

$$\mu \in \mathbb{R} \quad \sigma^2 \in \mathbb{R}^+$$

$$\Rightarrow \theta = (\mu, \sigma^2) \in (\mathbb{R} \times \mathbb{R}^+) \Rightarrow \Theta = \mathbb{R} \times \mathbb{R}^+$$

② $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} P(\lambda)$

$$\theta = \lambda \quad \Theta = \mathbb{R}^+$$

We discuss three methods of point estimation

(i) Unbiased estimation

(ii) Maximum likelihood estimation

(iii) Method of moment estimation

Unbiased Estimation Method

Def:- Let $X_1, \dots, X_n$ be a random sample from a popⁿ whose prob. distⁿ is given by $f_X(n, \theta)$, $\theta \in \Theta$ is the unknown parameter. An estimator $T(\underline{X})$, $\underline{X} = X_1, X_2, \dots, X_n$ is said to be unbiased for $g(\theta)$ if

$$E(T(\underline{X})) = g(\theta) \quad \forall \theta \in \Theta$$

Bias:- If $E(T(\underline{X})) = g(\theta) + b(\theta)$, then $b(\theta)$ is called the bias of estimator $T(\underline{X})$.

If $b(\theta) > 0 \rightarrow$ over estimate

If $b(\theta) < 0 \rightarrow$ underestimate

Ex:- ① Suppose $X \sim \text{Bin}(n, p)$ where n is known and $p, 0 \leq p \leq 1$ is unknown parameter. Find the estimator of p .

$$1 - \frac{s^2}{\bar{x}}$$

Ans:-

$$E(T(X)) = p$$

① Find unbiased estimator of p .

② " " " of p^2 .

③ " " " of $\text{var}(X) = np - np^2$

$$\textcircled{1} \quad T(X) = \frac{X}{n}$$

$$X \sim \text{Bin}(n, p)$$

$$E(X) = np$$

$$E(T(X)) = E\left(\frac{X}{n}\right) = \frac{1}{n} E(X) = \frac{np}{n} = p$$

$$\textcircled{2} \quad E(X(X-1)) = n(n-1)p^2$$

$$\Rightarrow E\left(\frac{X(X-1)}{n(n-1)}\right) = p^2$$

$\Rightarrow T(X) = \frac{X(X-1)}{n(n-1)}$ is unbiased estimator for p^2 .

$\textcircled{3} \quad T(X) = \frac{X(n-X)}{n-1}$ is unbiased estimator for $np - np^2$.

Ex2- $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2), \theta = (\mu, \sigma^2)$

Ans:- $\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}, E(\bar{X}) = \frac{n\mu}{n} = \mu$

$\Rightarrow \bar{X}$ is unbiased estimator of μ .

Similarly, $E(S^2) = \sigma^2$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

$\Rightarrow$ sample variance is unbiased est. of σ^2 .

Note:- ① Let $X_1, X_2, \dots, X_n$ be a random sample from a popⁿ with prob. distⁿ $f_X(x, \theta)$. If $E(X_i)$ exists, $i=1, 2, \dots, n$, then sample mean $\bar{X}$ is always unbiased for θ , where θ is the popⁿ mean. i.e. $E(\bar{X}) = \theta$

② If $E(X_i^2)$ exist or $\text{var}(X_i)$ exist, then sample variance is unbiased estimator for popⁿ variance say σ^2

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \quad \bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

$$E(S^2) = \frac{1}{n-1} E \left(\sum_{i=1}^n X_i^2 - n\bar{X}^2 \right)$$

$$\begin{aligned} & \sum (X_i - \bar{X})^2 \\ &= \sum X_i^2 - 2 \sum X_i \bar{X} + n\bar{X}^2 \\ &= \sum X_i^2 - 2n\bar{X}^2 + n\bar{X}^2 \\ &= \sum X_i^2 - n\bar{X}^2 \end{aligned}$$

$$\begin{aligned} E(X_i) &= \mu, E(\bar{X}) = \mu, \text{Var}(\bar{X}) = \sigma^2/n \\ E(\bar{X}^2) &= \mu^2 + \sigma^2/n \end{aligned}$$

$$\begin{aligned} &= \frac{1}{n-1} n(\mu^2 + \sigma^2) - \frac{n}{n-1} (\mu^2 + \sigma^2/n) \\ &= \frac{(n-1)\sigma^2}{n-1} = \sigma^2 \Rightarrow E(S^2) = \sigma^2 \end{aligned}$$

Ex 1:- $X_1, X_2, \dots, X_n \sim P(\lambda)$. find the unbiased estimator of λ .

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

$$E(X_i) = \lambda$$

$$E(\bar{X}) = \lambda$$

$$\text{Now, } E(X_1) = \lambda, E(X_2) = \lambda, \dots, E(X_n) = \lambda$$

All $X_i, i=1, 2, \dots, n$ are unbiased est. for λ .

$$E\left(\frac{X_1 + X_2}{2}\right) = \lambda \Rightarrow \text{We can have many unbiased estimators.}$$

Ex:- $X \sim \text{Bin}(n, p)$, $n \rightarrow \text{known}$
 $p \rightarrow \text{unknown}$

Let $T(x)$ be an unbiased est. of p^{n+1} .

$$\text{i.e. } E\{T(x)\} = p^{n+1}$$

$$\Rightarrow \underbrace{\sum_{x=0}^n T(x) \binom{n}{x} p^x (1-p)^{n-x}}_{\text{poly. of degree } n} = \underbrace{p^{n+1}}_{\text{poly. of degree } n+1} \text{---} \textcircled{1}$$

$\Rightarrow \textcircled{1}$ has no solⁿ

$\Rightarrow$ unbiased estimator may not exist.